OCR and Its Impact on Retrieval-Augmented Generation

Evaluation corpus document prepared for the Glyph RAG pipeline evaluation.

1. Overview

Retrieval-Augmented Generation (RAG) combines information retrieval with language generation. A user query is first matched against an external knowledge base, and the retrieved evidence is then supplied to a language model so that the answer can be grounded in external information. This makes the quality of the knowledge base an important part of the overall system.

2. Why Document Parsing Matters

A large amount of real-world information is stored in unstructured PDF documents. When PDFs contain scanned pages, tables, formulas, or complex layouts, optical character recognition (OCR) is commonly used to convert visual content into structured text. OCR is not perfect, however, and errors introduced during parsing can affect the quality of the downstream knowledge base.

3. OHRBench

OHRBench was introduced as a benchmark for studying the cascading impact of OCR on RAG systems. It contains 350 unstructured PDF documents from six application domains: textbook, manual, academic paper, newspaper, finance, and law. The benchmark contains 4,598 question-answer pairs derived from document evidence. The evidence sources include plain text, formulas, and tables, with short-text, judgment, numeric, and formula answer formats.

4. Types of OCR Noise

The study separates OCR noise into two main categories. Semantic Noise changes the meaning or content of extracted information through errors such as misspellings, misrecognized formula symbols, and incorrect table content or structure. Formatting Noise changes presentation or representation, including whitespace, markup, reading order, or formatting commands, without necessarily changing the underlying semantics.

5. Benchmark Construction

The benchmark begins with PDF documents and human-verified ground-truth structured data. Questions and answers are generated from evidence on individual document pages, including multimodal elements such as text, tables, and formulas. Quality control removes questions that can be answered without retrieval and eliminates ambiguous questions. Each question is designed to include a specific entity to reduce ambiguity.

6. Evaluation Method

The retrieval stage is evaluated using evidence inclusion measured with Longest Common Subsequence (LCS), reported as LCS@1. Generation quality is evaluated using Exact Match and F1-score. The study compares dense BGE-M3 retrieval with sparse BM25 retrieval, and evaluates Qwen2-7B-Instruct, Llama-3.1-8B-Instruct, and GPT-4o for generation.

7. Findings on Semantic Noise

Semantic Noise has a strong and consistent effect on RAG. Increasing semantic perturbation from mild to severe causes substantial performance degradation across retrieval and generation. The study reports that most retrievers and language models experience a decline approaching 50% in some settings. Semantic errors are especially damaging because they alter the information that the retriever and language model need to understand.

8. Findings on Formatting Noise

Formatting Noise has a more variable effect. It affects retrievers and language models differently and is particularly important for questions involving multimodal document elements. Formula and table questions can be sensitive to formatting changes. The study reports that GPT-4o is comparatively resilient to formatting noise, while some smaller models show larger performance reductions.

9. OCR and End-to-End RAG

OCR errors can propagate through the RAG pipeline. A parsing error can alter the knowledge base, reduce retrieval quality, and then affect the evidence available to the language model. The study therefore treats document parsing as an upstream component whose errors can cascade into retrieval, generation, and overall system performance.

10. Implications for RAG Applications

The results suggest that improving the retriever or language model alone is not enough when the knowledge base is built from noisy OCR. RAG systems that process complex PDFs should consider document parsing quality as part of the evaluation pipeline. The study also discusses Vision-Language Models (VLMs) as a possible alternative or complement to OCR. Combining visual input with OCR text was reported as a promising direction, although image-only input did not reach the performance of OCR text by itself.